

Homework 6

*Instructor: Henry Lam***Due Date:** Nov 3, 2025**Problem 1** For $Unif(0, 1)$ random variables $U_1, U_2, \dots$ define

$$N_1 = \text{Minimum} \left\{ n : \sum_{i=1}^n U_i > 1 \right\}.$$

That is, N_1 is equal to the number of random numbers that must be summed to exceed 1. Write a pseudo-code for generating N_1 . Give a point estimate and a 95% confidence interval for $\mathbb{E}[N_1]$, by generating 100, 1000 and 10,000 copies of N_1 respectively on a computer.

Do the same for N_2 , the number of random numbers that must be summed to exceed 2. That is,

$$N_2 = \text{Minimum} \left\{ n : \sum_{i=1}^n U_i > 2 \right\},$$

From your results, do you think $E[N_2]$ increases from $E[N_1]$? Explain briefly.

Problem 2 Suppose we estimate $\alpha = \mathbb{E}[X]$ via

$$\hat{\alpha}_n = \frac{1}{n} \sum_{j=1}^n X_j,$$

where the X_j 's are i.i.d. simulation replications of X . Assume that $\text{Var}(X) = 2$. Find the smallest value n such that $\hat{\alpha}_n$ is within 0.05 units of α with 95% confidence. Briefly justify your answer.

Problem 3 To estimate θ , we generate 30 *i.i.d.* replications each having mean θ . Suppose the obtained values are 102, 112, 131, 107, 114, 95, 133, 145, 139, 117, 93, 111, 124, 122, 136, 141, 119, 122, 151, 143, 130, 159, 92, 120, 114, 99, 135, 141, 125, 93.

- Find a 95% confidence interval for θ .
- How many additional replications do you think we will have to generate if we want to be 99% certain that our final estimate of θ is correct to within ± 0.5 ?

Problem 4 Let $X_1, \dots, X_{40}$ be *i.i.d.* copies of random variable X generated from a computer routine. Let

$$p = \mathbb{P}((X - 1)^2 < 30)$$

Give a point estimate and a 95% confidence interval of p if, among all the values of the X_i , 2 appears 11 times, 4 appears 5 times, 5 appears 10 times, 6 appears 2 times, 8 appears 7 times, 9 appears 4 times and 24 appears 1 time.

Problem 5 Suppose that the following are *i.i.d.* copies of random variable X generated from a computer routine: 5, 4, 9, 6, 21, 12, 7, 14, 17, 11, 20, 7, 10, 21, 15, 26, 9, 13, 8, 6. We want to estimate $\mathbb{E}[X^2]$.

- How many additional replications will we need to generate if we want to be 99% certain that our estimate has a margin of error 2?
- What assumptions have you made in part (a)?